Python Programming Fundamentals

List Methods and Loops

Programming Fundamentals Team

SETU

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Outline

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1. Adding Items

```python
shopping = []

shopping.append("milk")
shopping.append("bread")
shopping.append("eggs")

print(shopping)
```

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```
shopping = []

shopping.append("milk")
shopping.append("bread")
shopping.append("eggs")

print(shopping)
```

`append()` adds an item to the end of the list.

Outline

2. Removing Items

```python
shopping = ["milk", "bread", "eggs"]

shopping.remove("bread")
last_item = shopping.pop()

print(shopping)
print(last_item)
```

2. Removing Items

```
shopping = ["milk", "bread", "eggs"]

shopping.remove("bread")
last_item = shopping.pop()

print(shopping)
print(last_item)
```

- `remove(value)` removes a matching value
- `pop()` removes and returns an item

Outline

3. Looping Through a List

```python
names = ["Ava", "Ben", "Cara"]

for name in names:
    print(name)
```

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```python
names = ["Ava", "Ben", "Cara"]

for name in names:
    print(name)
```

This is the most Pythonic way when you need each item.

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4. Looping With Indexes

Sometimes you need positions too.

```python
names = ["Ava", "Ben", "Cara"]

for i in range(len(names)):
    print(i, names[i])
```

4. Looping With Indexes

Sometimes you need positions too.

```
names = ["Ava", "Ben", "Cara"]

for i in range(len(names)):
    print(i, names[i])
```

Python also has `enumerate()`:

```
for i, name in enumerate(names):
    print(i, name)
```

Outline

5. Processing Numeric Lists

```
grades = [55, 72, 81, 90, 66]

total = 0
for grade in grades:
    total = total + grade

average = total / len(grades)
print(average)
```

5. Processing Numeric Lists

```python
grades = [55, 72, 81, 90, 66]

total = 0
for grade in grades:
    total = total + grade

average = total / len(grades)
print(average)
```

This is the start of data processing.

Thanks for Watching - Any questions?